Business Requirement Document (BRD)

Project Name: Project Dalupotha (Factory Database & Supply Chain System) **Document Version:** 1.0

1. Executive Summary

Project Dalupotha aims to digitize and standardize the tea leaf collection and factory service management process. The system will bridge the gap between remote Small Holders, mobile Transport Agents, and central factory management. By introducing automated weight capture, GPS tracking, and a centralized portal for financial and agricultural services, the solution will eliminate manual data entry errors, ensure transparent supplier payments, and optimize logistical efficiency.

2. Business Objectives

- **Automate Data Capture:** Eliminate manual weighment logs by integrating IoT-enabled hanging scales via Bluetooth with mobile devices.
- **Enhance Traceability:** Mandate GPS location tagging for all Small Holder registrations and real-time tracking for Transport Agents during collection routes.
- **Centralize Operations:** Provide a unified digital interface to manage advances, balance payments, debts, and inventory (fertilizer, leaf bags).
- **Improve Supplier Experience:** Grant Small Holders direct system access (via OTP) to view their supply history, financial standing, and request services seamlessly.

3. Scope of Work

In-Scope:

- Small Holder registration module with GPS capture and OTP validation.
- Offline-first mobile application for Transport Agents to record collections and sync data.
- Management dashboards for factory, office, and store staff.
- Service request workflows for advances, fertilizers, transport, and machinery renting.
- Automated calculation logic for quality deductions, debts, and final balance payments.

Out-of-Scope:

- Direct control of internal factory manufacturing machinery (e.g., drying/rolling machines).

- Payroll processing for independent Small Holders (handled as supplier payouts, not internal HR).

4. User Roles and Access Profiles

The system caters to multiple stakeholders with distinct permission levels:

- **Small Holder:** Service receiver. Access via Username/Password and OTP. Can view histories and make requests.
- **Transport Agent (TA):** Executes field transport. Records leaf weight, syncs GPS, and manages on-field advance/fertilizer requests with permission.
- **Store Keeper (SK):** Manages physical inventory (Fertilizer stock, Leaf Bag stock).
- **Extension Officer (EXT) & Extension Manager (MG):** Monitors supplier relationships, manages advances, balance payments, debts, and advisory requests.
- **Office Staff (ST):** Handles the Brought Leaf Register, processes advanced/balance payments, and local sale tea tracking.
- **Factory Staff (FT):** Logs the quality of leaf, green leaf weights at the factory gate, and processes weight deductions.
- **Manager (MG):** Full oversight. Monitors dashboards, provides approvals (e.g., machine rent requests), and distributes TRI Circulars.

5. Functional Requirements

5.1 Registration & Authentication

- **REQ-1.1:** The system shall require OTP verification for Small Holder registration.
- **REQ-1.2:** Supplier profiles must capture: Name, Passbook No., Address, Land Name, GPS Location, In Charge No., and Arcs.

5.2 Field Collection & Transport (Mobile Module)

- **REQ-2.1:** The TA mobile interface shall connect to Bluetooth scales to pull exact Green Leaf Weights automatically.
- **REQ-2.2:** The system must capture the GPS coordinates at the exact moment of leaf collection.
- **REQ-2.3:** Transport Agents shall have access to view Green Leaf Supply History for verification purposes.

- **REQ-2.4:** The mobile application must operate completely offline in remote estates and automatically synchronize with the central database upon network availability.

5.3 Financial & Accounting Operations

- **REQ-3.1: Advances:** Workflows for Advance Requests and Advance History tracking. Accessible to EXT, MG, Office, SK, TA (with in-charge permission), and Small Holders.
- **REQ-3.2: Debts:** Automated tracking of outstanding debts visible to Small Holders, EXT, MG, TA, and SK.
- **REQ-3.3: Balance Payments:** Final payout calculations factoring in supplied leaf weight, quality deductions, debts, and advances. Visible to Small Holders, EXT, and MG.

5.4 Inventory & Factory Services (Stores & Office)

- **REQ-4.1: Stores Management:** Modules to track Leaf Bag Stock and Fertilizer Stock.
- **REQ-4.2: Fertilizer Requests:** Workflow originating from Small Holders or TAs, routed to the Office/SK for approval and dispatch.
- **REQ-4.3: Machinery Renting:** System to handle Machine Rent Requests, requiring Management approval.
- **REQ-4.4: Factory Logging:** Interface for Factory Staff to log the Brought Leaf Register, assess the Quality of Leaf, and apply necessary weight Deductions.

5.5 Communication & Dashboards

- **REQ-5.1:** A centralized dashboard displaying operational metrics for Management.
- **REQ-5.2:** Mechanism to distribute TRI (Tea Research Institute) Circulars to all system users.
- **REQ-5.3:** A portal for Small Holders to submit Advisory Requests and manage general Correspondences.

6. System Architecture & Integration Strategy

To maintain a high degree of scalability and modularity, the Dalupotha system will be structured as a plug-and-play architectural component within a broader ERP framework.

- **Module Integration:** The financial workflows (Balance Payments, Advance History, Debts) will integrate directly with the **Ginum** accounting module to ensure seamless ledger updates and automated payout processing. User role administration and factory staff tracking will interface with the **PirisaHR** module.

- **Data Synchronization:** Real-time data routing and offline-sync resolution for the Transport Agent mobile application will be handled utilizing **Supabase** as the backend database layer.
- **Infrastructure:** The backend services and web dashboards will be deployed across robust cloud environments such as **Digital Ocean**, **Oracle Cloud Infrastructure (OCI)**, or **Microsoft Azure** to guarantee continuous uptime, especially during peak harvesting seasons when system loads are highest.

7. Assumptions & Constraints

- **Assumptions:** Industry-standard hanging scales purchased will have open Bluetooth APIs for integration. Small Holders possess mobile devices capable of receiving SMS OTPs.
- **Constraints:** Extreme weather conditions and poor connectivity in tea estates dictate that hardware must be ruggedized, and the system cannot rely on continuous server polling to execute core collection functions.